Multi-contact Optimization for Whole-body Dexterity

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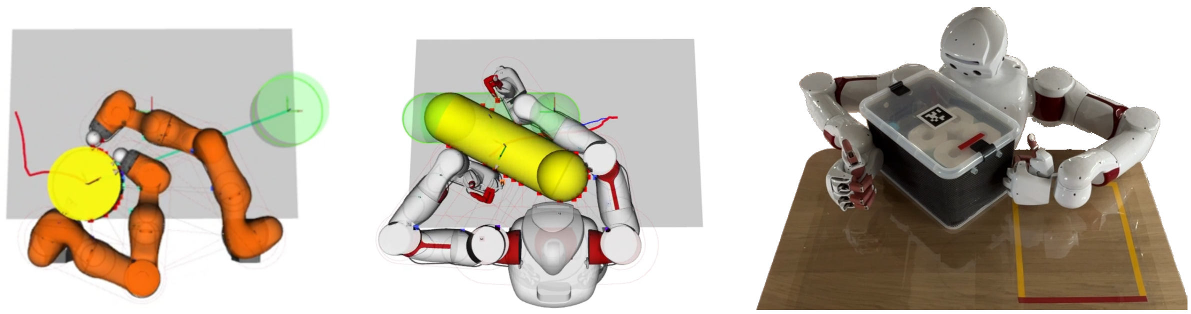


Fig. 1: Examples of autonomously planned contacting configurations in Multi-contact Whole-body Manipulation.

***Abstract*— Humans are able to use their whole body to dexterously manipulate objects even when their hands are unavailable. For robots, allowing the creation of contact anywhere on the body surface implies the computational challenge of deciding among infinitely many possibilities of contact locations. To deal with this, existing methods mainly rely on manual or random sampling to choose contacts on the robot. Instead, our approach is to enable continuous optimization of the contact location to improve the convergence of the manipulation planning and thus increase the whole-body dexterity of the robot. In our work, we proposed a cost design for contact optimization that effectively guides whole-body manipulation planning and relies on a closed-form formulation of proximity points between the robot and object surfaces. Our experiments demonstrate that our proposed method can solve problems unaddressed by existing methods, and achieves a 77% improvement in planning time over the state of the art. We also validate the suitability of our approach on real hardware through the whole-body manipulation of boxes by a humanoid robot. Research page.**

I. INTRODUCTION

A. Overview

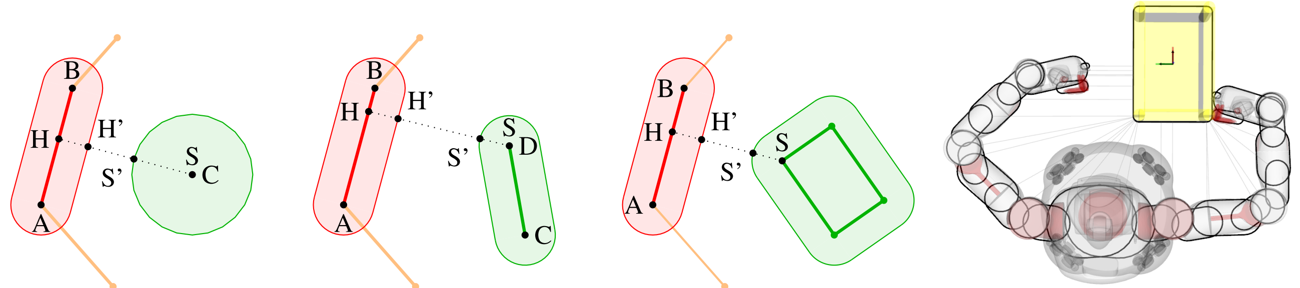
The autonomy of humanoid robots lies in their ability to automatically translate high-level commands into actionable plans. In an object manipulation task, a human should only need to specify the desired pose of the object, with the corresponding sequence of contacts and joint trajectory computed autonomously to accomplish the task. The task objective, along with the geometry and physical properties of both the robot and the object, will dictate different contact strategies. For instance, these may involve using the robot's torso as a contact surface to hold a heavy item while keeping a hand free to open a door. We refer to the problem of computing such motions by leveraging the robot's full contact surface as the Whole-body Multi-contact Manipulation (WMM) problem.

B. Approach and challenges

Planning contact-rich motions is a high-dimensional problem that requires discrete choices of contact modes (sticking, sliding, breaking) across infinitely many potential contact locations resulting in a combinatorial complexity that grows exponentially [1]. Without an efficient exploration strategy, it may take hours to find a feasible whole-body manipulation plan. Ideally, a system model would guide the search toward actions that directly support the manipulation goal, and [2], [3] show us how important contact optimization is to perform a contact-rich task. Given the continuous nature of contact surfaces, this search guide could be enabled with the formulation of the WMM planning as a continuous Nonlinear Programming Problem (NLP). However, two challenges remain to achieve this: (a) Finding a continuous differentiable representation of the proximity points on the object and robot surfaces that allows Contact Optimization (CO) in a NLP; (b) Finding an optimization cost design that guides the planner to avoid local minima and permit dexterous manipulation using an undefined number of contacts even on body parts with lower Degrees of Freedom (DOF), such as upper arms.

C. Related works

To deal with these challenges, the state-of-the-art relies on manual or random sampling of contact locations to break down the complexity of infinite possibilities of contact location on the surface of the robot [4], [5]. However, such sampling is unrealistic for covering large surfaces like the robot whole body. Several works proposed continuous representations for either the robot or object surfaces, but still require the sampling of contact on the other surface to work [6], [7], [8], or elude important aspects of the WMM [9], e.g. the sliding mode, making it impractical for hardware transfer. Approaches based on Linear Quadratic Regulator work well for local manipulation of soft objects [10] but is not efficient for rigid objects [11]. The contact optimization in the state-of-the-art model-based planning method consists in simplistic cost design without consideration for manipulability [5], [8] or on methods tailored for grasping rather than manipulating [4], hindering dexterous manipulation required



(a) Point Proximity (b) Segment Proximity (c) Polygon Proximity (d) Contact surface representation

Fig. 2: Representation of the contact surface for the robot and object. The robot contact surface is represented by a set of rounded segment patches (d). The object can be represented by a circle (a), a rounded segment (b) or a rounded polygon (c)(d). The collision avoidance between robot parts pairs is represented by two rounded segments (b).

for complex scenarios. The prediction of an horizon outcome can help guiding the manipulation [12], [13], [14], but to remain computationally efficient this method ignore the full kino-dynamic constraints of the robot that are essential for WMM.

Learning-based methods are restricted to systems without arm kino-dynamic constraints [15], with simplified geometries [16], require model-based planning input to converge [17] or human input to plan global contacts [18].

D. Contributions

In our works [1], [19], we investigated both representations of surfaces that enable CO and cost design to plan autonomously WMM through contacts. The representation requires a trade-off between the accuracy of the surface characterisation and its computational efficiency when integrated into a NLP, e.g. its differentiability and convexity. We first proposed a surface representation as a differentiable function using Gaussians [1] with a planning pipeline for WMM. It allowed to precisely describe the non-convex outline of the robot while integrating into a NLP with CO. Despite a clear performance improvement compared to the state-of-the-art sampling planner, we identified limitations when scaling this first planner to multi-contact WMM.

To cope with this, we proposed in our latest work [19] a representation based on convex primitive patches and a novel cost function to dexterously plan contacts with any parts of the robot that we explain in this paper. Our experiments show that our representation is accurate enough to permit transfer to real robot hardware with complex geometry. We demonstrate the time performance superiority of our planning, with a 77% improvement of the planning time on average and higher success rates over the state of the art WMM sampling-based planner [4]. We also highlight the benefit of our cost design over a baseline cost function [5] for avoiding local minima in complex WMM scenarios with 100% success rate. Our approach also profits from no post trajectory refinement needed, no task-dependent tuning, and an easy adaptation to new robot model through geometric parametrization.

E. Problem statement

The WMM planning problem we address in this work consists in finding a trajectory of robot actions $u^{(t)}$ that brings an object from an initial state q_u^{init} to a goal state q_u^{goal}, the robot starting in the state q_a^{init}.We assume the manipulation to be planar, the dynamics to be quasi-dynamics and the object to be convex. We also assume the planning to take into account all the kino-dynamic constrains of the system, namely the robot joint and torque limits, the robot self-collision avoidance, the object workspace boundaries and the dynamics of the different contact modes. See section III of our latest work [19] for details on the problem formulation.

II. DEXTEROUS PLANNING

A. Robot and object surface representations

In [19] we proposed a method to compute the proximity points between a robot surface and the object surface, which novelty is to be formulated as a differentiable closed-form. This representation allows us to formulate continuous optimization of contact location on both the robot and object surfaces. The robot is represented by a set of rounded segment patches pictured in Fig. 2d, and the object is represented either as a rounded point (Fig. 2a), segment (Fig. 2b) or convex-polygon (Fig. 2c). Details about the computation of the proximity points for different pairs of structure shapes can be found in the section IV.A of our latest work [19].

B. Pipeline

We explain below the main components of the pipeline of our proposed planner Fig. 3.

1) Global Planner: The role of the global planner is to ensure that the planning does not get stuck in local minimum and find a solution in most complicated scenarios. It is implemented as a Rapidly-exploring Random Tree (RRT) with rewiring of similar nodes, building a graph of explored system states by iterating through trajectories generated by the trajectory optimizer described next.

2) Trajectory optimizer: The trajectory optimizer receives from the global planner the nearest node q^{near} and the subgoal q^{samp}. It generates trajectories by first running a N_{cf} steps contact-free Placement Trajectory Optimization (TO),

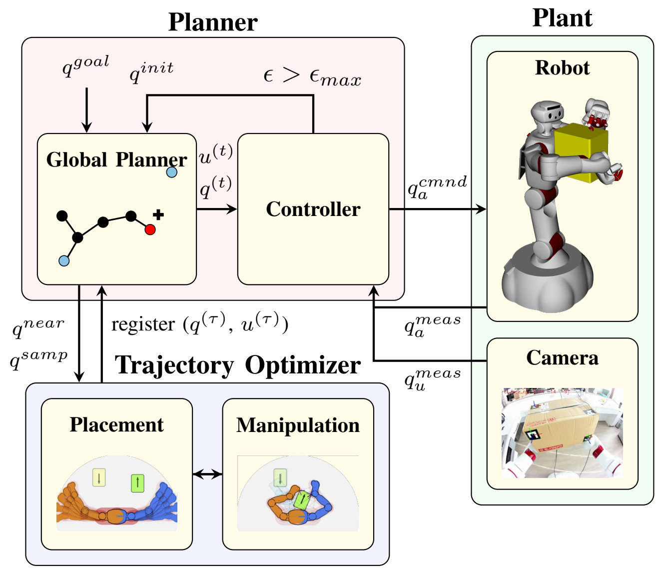


Fig. 3: Pipeline of the proposed planner

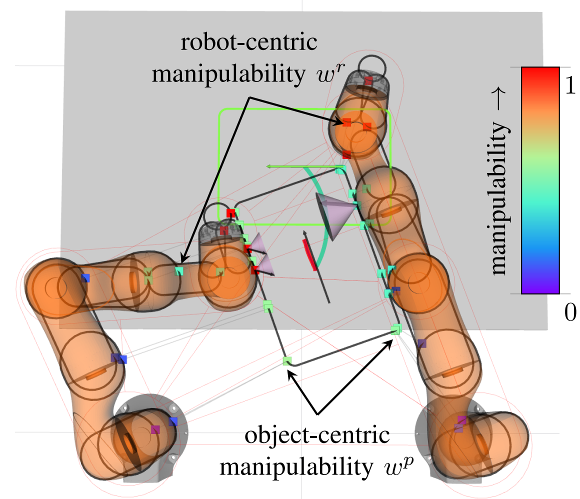


Fig. 4: Manipulability metrics in contact optimization.

which seeks contact with the object without applying any force, and then once a contact is found it switches to a N_{cr} step contact-rich Manipulation TO. This allows to run the crucial Placement phase with a longer horizon while keeping computationally reasonable. Here we describe the cost design used in both the Placement TO and Manipulation TO, but these TOs are also subject to kino-dynamic constraints detailed in [19].

Placement: The Placement TO seeks a robot configuration that contacts with the object surface while optimizing the contact locations to have good robot-centric and object-centric manipulability metrics. The optimization objective formulates as follows:

$$\min_{u} \quad \mathcal{G}_p(q^{N_{cf}}) + \mathcal{G}_r(q^{N_{cf}}) \tag{1a}$$

$$\text{with} \quad q^+ = q + \mathbf{M}^{-1}\mathbf{K}u \tag{1b}$$

$$\mathcal{G}_p(q) = -\beta_p \sum_{k\in\mathcal{C}} w_k^p(q) w_k^d(q) \tag{1c}$$

$$\mathcal{G}_r(q) = -\beta_r \sum_{k\in\mathcal{C}} w_k^r(q) w_k^d(q) \tag{1d}$$

$$w_k^d(q) = \ln\left(1 + \exp\left(-\beta_d d_k^r(q)\right)\right) \tag{1e}$$

$$w_k^p(q) = \left(1 - \left(\frac{\phi_k(q)}{\pi}\right)^2\right)\left\langle \frac{\tilde{f}_k^n(q)}{f_{lim}} \right\rangle_0^1 \tag{1f}$$

$$w_k^r(q) = \det\left(J_k^r(q) J_k^r(q)^\top + \beta_j I\right) \tag{1g}$$

$$\phi_k(q) = \arctan(\tilde{f}_k^t(q), \tilde{f}_k^n(q)) \tag{1h}$$

$$\tilde{f}_k^n(q) = J_{u,k}^n(q)(q_u - q_u^{goal}) \tag{1i}$$

$$\tilde{f}_k^t(q) = J_{u,k}^t(q)(q_u - q_u^{goal}), \tag{1j}$$

where d_k^r is the proximity distance between the object and the k^{th} patch of $\mathcal{B}_r$ representing the robot surface. β_d, β_p, β_r and β_j are tunable scalar gains. The costs $\mathcal{G}_p$ and $\mathcal{G}_r$ correspond respectively to whole-body object-centric and robot-centric manipulability metrics. Manipulability metrics for each contact candidate are illustrated in the example of Fig. 4.

w_k^p is an object-centric manipulability metric evaluated at the k^{th} contact and based on the angle of reference force $\tilde{f}_k$ to the contact normal. This metric first computes a reference force $\tilde{f}_k$ to apply at the k^{th} contact if it was active, based on the object state deviation $q_u - q_u^{goal}$, with $\tilde{f}_k^n$ and $\tilde{f}_k^t$ respectively the normal and tangential components of this reference force. The angle ϕ_k of the reference force to the contact normal is then used to evaluate the manipulability, with higher manipulability when the angle is close to 0 and lower one as it increases towards π.

w_k^r is a Yoshikawa's manipulability metric [20] evaluated at the k^{th} contact. We adapted it with a regularization term β_j to provide a strictly positive value even for contacts on links with a rank-deficient Jacobian matrix, otherwise the manipulability would be always zero for these links. This helps planning contacts with the different body parts of the robot based on their DOF.

w_k^d is an activation function weighting how close the k^{th} contact is to be active based on the distance d_k^r. We tested different activation functions such as linear or simple exponential, but these either led the robot towards contacts too fast without increasing much metrics w_k^p and w_k^r, or on the other hand failed at creating contacts, the robot moving continuously around the object without actually touching it.

Manipulation: The contact-rich Manipulation phase takes into account the deviation of the current object state q to the goal state q^{goal}. Its objective function formulates as

$$\min_{u,\nu} \quad \alpha\mathcal{G}_p(q^{N_{cr}}) + \alpha\mathcal{G}_r(q^{N_{cr}}) + \mathcal{G}_d(q^{N_{cr}}) \tag{2a}$$

$$\text{with} \quad \mathcal{G}_d(q) = (q - q^{goal})^\top W_d (q - q^{goal}), \tag{2b}$$

where α is a diagonal matrix of binary values encoding which of all the possible contact pairs between the robot and the object surface patches are currently active (1 for

Planning time		Scenario 1					Scenario 2					Scenario 3	Scenario 4
q_a^{init}	[deg]	-40.1, 0, 0, 40.1, 0, 0					0, 0, 0, 0, 0, 0					-45, 0, 0, -45, 0, 0	0, 0, 0, 0, 0, 0
q_u^{init}	[m, m, deg]	0.60, -0.20, 0.00					0.65, 0.00, 0.00					0.50, -0.55, 0.00	0.40, 0.00, 0.00
q_u^{goal}	[m, m, deg]	0.60, 0.20, θ					0.35, 0.00, θ					0.50, 0.55, -90.0	0.80, 0.00, 0.00
θ	[deg]	0	45	90	135	180	0	45	90	135	180	-	-
OURS	success[1] [%]	100	100	100	100	93	100	100	100	100	100	100	100
	mean [mn]	1.2	1.9	5.7	6.5	16.0	0.5	0.4	5.3	7.8	5.3	5.6	2.6
	min [mn]	1.0	1.7	3.4	4.6	6.0	0.3	0.4	4.6	4.5	3.1	5.2	2.3
	max [mn]	1.3	2.1	11.2	10.1	-	0.7	0.5	8.1	12.5	12.9	6.1	2.7
Baseline[2]	success[1] [%]	100	70	7	0	0	93	40	33	7	0	100	20
	mean [mn]	3.9	13.5	29.1	-	-	9.7	20.9	22.7	28.7	-	15.2	27.9
	min [mn]	0.3	0.6	5.9	-	-	0.4	0.3	0.5	2.7	-	8.2	12.1
	max [mn]	26.1	-	-	-	-	-	-	-	-	-	21.4	-

Fig. 5: Comparison of planning performance our method (OURS) and a Baseline[2]. [(1)] Planning time exceeding the timeout of 30mn is considered as failure. [(2)] Baseline for scenarios 1 and 2 is the sampling planner [4], and for scenarios 3 and 4 is our planner but with the optimization cost function from [5].

active contact, 0 for inactive). W_d is a diagonal matrix for weighting the position and rotation of the deviation between the current state and the goal state.

III. EXPERIMENTS

To highlight the merit of using our planning method for WMM, we report about two experiments we conducted: (A) a performance comparison between our planner and the state-of-the-art WMM sampling planner, demonstrating the advantage of using continuous optimization in WMM enabled by our recent contributions; (B) an ablation study that evaluates the performance of our planner with our proposed cost design and with another baseline cost design, to confirm the benefit of our cost design for avoiding local minimum and improving time performance. The setup consists in two KUKA iiwa 7 robots and a 0.14m radius cylinder object constrained to move on a plane. Additional information about these experiments and about hardware transfer can be found in our video and in the section V of our latest paper [19].

A. Continuous Contact Optimization VS Sampling

1) Protocol: We compare the performance of our planner with the sampling planner described in [4] as a baseline. We consider two scenarios 1 and 2 consisting in a translation with re-orientation to an increasing angle θ (see Fig. 5 for exact scenario conditions).

2) Results: The results for scenarios 1 and 2 shown in Fig. 5 empirically demonstrate that our method has several advantages compared to the baseline. The overall planning time is 77% faster in average across all re-orientations, which confirms our assumption that guiding the planning search with CO is beneficial. The planning time of our planner is also more consistent, and the success rate higher, with 100% success rate for most of the scenarios. Finally, our method implements all kino-dynamic constraints at each step of the planning, so no refinement of the final trajectory is required, unlike the baseline planner which requires a refinement that fails sometimes.

B. Evaluation of cost design

1) Protocol: In this experiment we assess the improvement of the contact optimization induced by our cost design on two more difficult scenarios 3 and 4. To evaluate this, we compare the planning performance of our planning pipeline with our cost design and with the baseline cost (based on distance of candidate contact location to the object centre) from [5]. The scenario 3 consists in moving a cylinder from one edge of the table to the other, which requires autonomous switching between single arm and dual-arm manipulation of the object, and contacts with various links of the robots. The scenario 4 is a forward pushing from a position difficult to reach for the end-effectors, encouraging the use of body parts with lower DOF.

2) Results: The results in Fig. 5 show that our cost design helps keeping planning time reasonable ($<$ 10mn) even on more complex scenarios. On the contrary, using the baseline costs requires more time to find a solution, and even fails at finding one within 30mn for the scenario 4, 80% of the time. The planner with our cost design is able to create better contacts that helps the robot to avoid local minima where the object is difficult to manipulate towards the goal, without relying on randomness of RRT, improving the repeatability of the planning at no cost.

IV. DISCUSSION

In our works [1], [19] we proposed means of conducting continuous optimization of contact locations on the robot and object surfaces for WMM. These contributions demonstrated the possibility to improve the whole-body dexterity of robots by reducing planning time and consistency, even in complex multi-contact scenarios. We only validated our method on planar WMM at the moment, but we plan to adapt our representation and cost design to 3D whole-body manipulation of more sophisticated objects in the near future. We also believe that better metrics to anticipate the horizon outcome of the manipulation could improve further the whole-body dexterity of robot manipulation planners using contact optimization.

REFERENCES

[1] V. Levé, J. Moura, N. Saito, S. Tonneau, and S. Vijayakumar, "Explicit contact optimization in whole-body contact-rich manipulation," in *IEEE-RAS 23rd Humanoids*, Nov. 2024, p. 451–458, link.

[2] M. Li, K. Hang, D. Kragic, and A. Billard, "Dexterous grasping under shape uncertainty," *Robotics and Autonomous Systems*, vol. 75, pp. 352–364, 2016.

[3] L. Yan, T. Stouraitis, J. Moura, W. Xu, M. Gienger, and S. Vijayakumar, "Impact-aware bimanual catching of large-momentum objects," *IEEE T-RO*, vol. 40, pp. 2543–2563, 2024.

[4] T. Pang, H. J. T. Suh, L. Yang, and R. Tedrake, "Global planning for contact-rich manipulation via local smoothing of quasi-dynamic contact models," *IEEE T-RO*, vol. 39, no. 6, pp. 4691–4711, 2023.

[5] W. Jin, "Complementarity-free multi-contact modeling and optimization for dexterous manipulation," *RSS*, 2025.

[6] Y. Li, Y. Zhang, A. Razmjoo, and S. Calinon, "Representing robot geometry as distance fields: Applications to whole-body manipulation," in *IEEE ICRA*, 2024, pp. 15 351–15 357.

[7] M. Zhang, D. K. Jha, A. U. Raghunathan, and K. Hauser, "Simultaneous trajectory optimization and contact selection for contact-rich manipulation with high-fidelity geometry," *IEEE T-RO*, vol. 41, pp. 2677–2690, 2025.

[8] W. Yang and W. Jin, "Contactsdf: Signed distance functions as multi-contact models for dexterous manipulation," *IEEE RA-L*, vol. 10, no. 5, pp. 4212–4219, 2025.

[9] I. Mordatch, Z. Popović, and E. Todorov, "Contact-invariant optimization for hand manipulation," in *ACM SIGGRAPH/EG SCA*, 2012, p. 137–144.

[10] V. Kurtz and H. Lin, "Contact-implicit trajectory optimization with hydroelastic contact and ilqr," in *IEEE/RSJ IROS*, 2022, pp. 8829–8834.

[11] Y. Shirai, T. Zhao, H. T. Suh, H. Zhu, X. Ni, J. Wang, M. Simchowitz, and T. Pang, "Is linear feedback on smoothed dynamics sufficient for stabilizing contact-rich plans?" in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 926–11 932.

[12] J. Moura, T. Stouraitis, and S. Vijayakumar, "Non-prehensile planar manipulation via trajectory optimization with complementarity constraints," in *IEEE ICRA*, 2022, pp. 970–976.

[13] C. Pezzato, C. Salmi, E. Trevisan, M. Spahn, J. Alonso-Mora, and C. Hernández Corbato, "Sampling-based model predictive control leveraging parallelizable physics simulations," *IEEE RAL*, vol. 10, no. 3, pp. 2750–2757, 2025.

[14] A. Aydinoglu, A. Wei, W.-C. Huang, and M. Posa, "Consensus complementarity control for multicontact mpc," *IEEE T-RO*, vol. 40, pp. 3879–3896, 2024.

[15] S. Jeon, M. Jung, S. Choi, B. Kim, and J. Hwangbo, "Learning whole-body manipulation for quadrupedal robot," 2023.

[16] T. Chen, J. Xu, and P. Agrawal, "A system for general in-hand object re-orientation," in *CoRL*, vol. 164. PMLR, 2022, pp. 297–307.

[17] M. Zhang, J. Barreiros, and A. O. Onol, "Plan-guided reinforcement learning for whole-body manipulation," 2023.

[18] L. Yang, H. T. Suh, T. Zhao, B. P. Graesdal, T. Kelestemur, J. Wang, T. Pang, and R. Tedrake, "Physics-Driven Data Generation for Contact-Rich Manipulation via Trajectory Optimization," in *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, June 2025.

[19] V. Levé, J. Moura, S. Fujita, T. Miyake, S. Tonneau, and S. Vijayakumar, "Scaling whole-body multi-contact manipulation with contact optimization," 2025, accepted at IEEE-RAS 24th Humanoids. link.

[20] T. Yoshikawa, "Manipulability of robotic mechanisms," *Int. J. Robot. Res.*, vol. 4, no. 2, pp. 3–9, 1985.